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Creative Destruction and the Economic Impact of Artificial Intelligence

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Artificial intelligence is advancing quickly and diffusing across sectors: what once belonged to experimental settings is now embedded in core workflows - planning and forecasting, production, customer interaction, research, and creative work. Today, around 88 percent of firms globally report using AI in at least one business function, suggesting that adoption is no longer exceptional but increasingly routine. At the same time, the global AI market is projected to reach nearly \$758 billion by 2026, and more than nine out of ten leading firms report active investment in AI, indicating that AI is becoming a standard component of competitive strategy rather than a peripheral technology (Naik, 2025).

The question is no longer whether AI will matter, but how it will reshape economic structures: which firms gain advantages, which business models weaken, how value chains reorganize, and where social and political pressure points appear. In sectors such as healthcare, for example, this transformation is already visible - the global AI healthcare market is expected to exceed \$110 billion by 2030, driven by adoption in diagnostics, clinical workflows, and administrative automation (Shaikh, 2025). The scale and speed of this diffusion matter because they suggest that AI is not simply improving existing processes - but beginning to alter how economic activity is organized, which capabilities remain valuable, and who is likely to benefit from technological change.

This paper approaches the question through a structural rather than a purely technological lens - AI is treated not as an isolated breakthrough, but as a force that operates through reallocation and replacement, as argued by Schumpeter (1942) and formalized by Aghion and Howitt (1992). The central framework is creative destruction, which links innovation to productivity growth while also explaining why growth is rarely smooth: progress arrives through displacement and conflict, through the breakdown of established arrangements and the

construction of new ones, rather than through gradual adjustment around a stable equilibrium. Building on this logic, the paper synthesizes four core contributions operating at different levels: Schumpeter (1942) frames capitalism as an innovation-driven system in which competition rewrites market structures from within; Christensen (1997) shows at the firm level why displacement is systematic rather than accidental; Aghion and Howitt (1992) formalize innovation as sequential replacement that raises growth and volatility while creating wedges between private and social returns; and Bresnahan and Trajtenberg (1995) extend the analysis to the system level, showing that broad-based growth emerges only when diffusion, complementarities, and institutional adaptation move together.

These contributions explain why AI can generate strong disruptive pressure, but they also expose a gap in how transformation unfolds in practice. Disruption on its own does not explain why technological change translates into sustained development in some contexts while stalling or fragmenting in others. To address this gap, the paper turns to insights from the Swedish Growth School, which shift attention away from disruption itself and toward the conditions under which transformation is completed. In this perspective, Dahmén (1988) highlights structural tensions and missing complements, Carlsson and Stankiewicz (1991) situate these tensions within broader technological systems, Acs and Sanders (2013) emphasize entrepreneurship as the mechanism through which knowledge is converted into economic activity, and Acs and Audretsch (1987) show how market structure conditions where and by whom innovation occurs.

Literature Review

Part I: Creative Destruction

Schumpeter's starting point is a redefinition of competition. In the standard static view, firms compete mainly through prices within relatively stable markets. Schumpeter argues that this misses the essence of capitalism. The decisive form of competition operates through innovation - new products, new production methods, new sources of supply, new markets, and new forms of industrial organization - that does not take place within a fixed structure but repeatedly reshapes it from within Schumpeter (1942). Capitalism, in this sense, is not a system tending toward equilibrium with occasional disturbances. It is a moving frontier in which the basis of competition is continuously rewritten. For this reason, creative destruction should not be understood as mere technical substitution, where one tool replaces another while the surrounding system remains intact. As Schumpeter explains:

“It must be seen in its role in the perennial gale of creative destruction; it cannot be understood irrespective of it or, in fact, on the hypothesis that there is a perennial lull. In other words, the problem that is usually being visualized is how capitalism administers existing structures, whereas the relevant problem is how it creates and destroys them.” (Schumpeter, 1942, p. 84)

In Schumpeter's account, this is not a failure of capitalism, but its defining mechanism - a continual reshaping of the economic landscape through replacement rather than smooth adaptation. This implies that economic progress and social instability are inseparable. Creative destruction destroys rents, jobs, and identities and therefore generates resistance. Over time,

capitalism may falter not because it becomes inefficient, but because social and political reactions to disruption harden into institutional constraints. In the context of AI, this is not just about replacing one machine with another - it is about redistributing tasks within jobs, reshaping workflows, and reallocating labor across occupations: surveys of U.S. businesses show that AI adoption is already changing how work is done, particularly in the allocation of tasks that machines can perform more efficiently than humans, with implications for employment across sectors (Press, 2025). At a broader level, recent economic forecasts suggest that a large share of work hours could be affected by AI-driven automation - for example, some analysts project that up to 25% of all work hours could be automated in coming years, altering the nature of many jobs rather than eliminating them entirely, while potentially boosting productivity in the process (Times of India Tech Desk, 2026). These patterns make clear that obsolescence under AI becomes structural rather than accidental: a firm can be efficiently managed within an established model and still lose relevance as its tasks, capabilities, and roles are reorganized around AI-enabled systems.

While Schumpeter provides a powerful conceptual account of capitalist development, his analysis remains largely descriptive. To understand how creative destruction translates into sustained economic growth - and why this growth is inherently unstable - it is necessary to formalize the mechanism linking innovation, incentives, and replacement. Here Aghion and Howitt formalize Schumpeter's insights come in to play by modeling economic growth as a process of sequential technological replacement. In their framework, growth does not arise from smooth accumulation of capital or labor but from successive innovations that improve productivity while rendering existing technologies obsolete. Destruction is not an external cost of progress; it is embedded in the growth process itself.

Innovation is motivated by temporary monopoly rents. Firms invest in R&D because successful innovation grants time-limited market power and profits. However, the model introduces a critical intertemporal tension. Expectations of future innovation reduce incentives to innovate today because future breakthroughs shorten the effective lifespan of monopoly rents. As Aghion & Howitt (1992, p. 331) explain, “Each innovation is an act of creation aimed at capturing monopoly rents. But it also destroys the monopoly rents that motivated the previous creation” This creative destruction effect captures how rapid technological turnover can compress private payoffs and reshape investment horizons. The model also highlights uncertainty as a structural feature of innovation-led growth. Output follows a stochastic process: higher innovation intensity raises average growth but also increases volatility. Growth and instability therefore move together. Innovation-driven development is inherently turbulent, generating macroeconomic and labor-market instability rather than converging smoothly to equilibrium (Aghion & Howitt, 1992).

Finally, Aghion and Howitt show that decentralized markets do not necessarily produce socially optimal innovation outcomes. Although innovation is driven by the promise of temporary monopoly rents, firms make decisions based on private payoffs rather than economy-wide effects. When new innovations displace existing ones, entrants capture profits from replacement, but they do not take into account the disruption of existing productive structures and the adjustment costs imposed on workers and firms. As a result, innovation incentives can become misaligned with social welfare - pushing innovation to occur either too quickly, too slowly, or in directions that amplify instability rather than long-run development. In this framework, growth through creative destruction is powerful, but not automatically efficient,

which opens space for institutions and policy to shape outcomes rather than block innovation altogether (Aghion & Howitt, 1992).

While the Aghion–Howitt model explains how creative destruction generates growth and volatility at the macro level, it does not explain why disruption often appears as the failure of strong incumbent firms. For this, a firm-level perspective is required. In advanced capitalism, Schumpeter argues, innovation increasingly takes place within large firms with internal R&D capabilities. Innovation becomes routinized, relying less on individual entrepreneurs and more on organized research, managerial coordination, and capital-intensive infrastructure. This organizational shift changes how innovation is produced and governed and, in doing so, reshapes the political economy of technological change. As Schumpeter notes, “These units not only arise in the process of creative destruction and function in a way entirely different from the static schema, but in many cases of decisive importance they provide the necessary form for the achievement. They largely create what they exploit” (Schumpeter, 1942, p. 101).

Christensen builds directly on this firm-level reality by explaining how creative destruction unfolds inside organizations. The Innovator’s Dilemma shows that leading firms fail not because they are poorly managed, but because they are well managed - they rationally allocate resources toward profitable customers, proven improvements, and existing business models. Over time, these choices create forms of path dependence and technology lock-in - organizational routines and entrenched commitments that make certain technological directions harder to adopt, even when they are promising Christensen (1997). This logic is central to understanding why displacement under creative destruction is systematic rather than accidental. Firms do not resist new technologies because they fail to recognize them, but because their internal incentive structures reward sustaining innovations that reinforce existing trajectories.

Christensen captures this problem succinctly when he notes that “markets that do not exist cannot be analyzed: suppliers and customers must discover them together,” and that the appropriate response to disruptive technological change is therefore “learning and discovery rather than plans for execution” (Christensen, 1997, p. 222). When firms rely on established planning and forecasting routines, they systematically underinvest in technologies whose markets are uncertain, small, or initially unprofitable.

Christensen further clarifies this mechanism through his distinction between sustaining and disruptive innovations. Disruptive technologies typically underperform on mainstream performance metrics and often appear inferior when judged by incumbent standards. They are frequently cheaper, simpler, or targeted at niche or low-end segments. Because these markets seem unattractive relative to the firm’s core business, organizations deprioritize them. Resource allocation processes then institutionalize this bias, channeling capital and talent toward sustaining innovations that improve existing products and serve existing customers, while disruptive paths remain underexplored. Displacement therefore occurs not through direct technological competition, but through market redefinition. Disruptors introduce new value propositions and attract new users who are not well served by incumbent offerings. Over time, performance in these new trajectories improves faster than customer requirements increase, and displacement can be rapid once the new basis of competition becomes dominant. Firm failure, in this sense, is not a surprise - it is the predictable outcome of organizational routines, incentive structures, and internal metrics that remain aligned with an outdated definition of performance. (Christensen, 1997)

Christensen’s analysis also explains why creative destruction unfolds unevenly across firms. Outcomes depend on adaptive capacity - specifically, on whether firms can create

organizational spaces that allow disruptive experimentation without being dominated by the core business. The barrier is often not access to technology, but misalignment between organizational values, incentive systems, and the emerging basis of competition. Where such spaces exist, firms can experiment, learn, and reposition themselves. Where they do not, even technologically sophisticated organizations remain vulnerable to displacement (Christensen, 1997). This framework is especially relevant in the context of AI. Recent scholarship points out that harmful path dependencies can emerge when early design and development decisions become embedded in models, data infrastructures, and organizational systems, making later alternatives difficult to adopt without deliberate intervention (Chiodo, Müller & Sienknecht, 2024). Choices that are optimal under existing incentives - such as optimizing models for short-term efficiency, scale, or narrow performance benchmarks - can become traps once AI changes what performance means at the system level. Firms may then struggle to adapt even when new AI-enabled approaches clearly advance.

Although Christensen clarifies how firms are displaced, the economic impact of certain technologies extends beyond firm-level dynamics. Some technologies reshape entire sectors and production systems, requiring a broader analytical lens. Bresnahan and Trajtenberg introduce the concept of general purpose technologies (GPT) - technologies that drive long growth waves by transforming production across many sectors at once. Historically, steam power, electricity, and semiconductors played this role (Bresnahan & Trajtenberg, 1995). AI fits naturally into this category: its applications are pervasive across industries, its performance improves continuously through data and learning, and its economic impact depends heavily on complementarities with downstream innovations - new applications, organizational changes, and institutional adaptation - rather than on standalone adoption.

The key insight of the GPT framework is that such technologies function as enabling technologies rather than final solutions. Their productivity impact does not arise from adoption alone, but from complementary investments - new applications, organizational redesign, sector-specific process innovations, and new forms of coordination. Bresnahan and Trajtenberg emphasize that GPTs “open up new opportunities rather than offering complete, final solutions,” and that their economic effects depend on innovational complementarities, whereby innovation in the core technology raises the productivity of downstream R&D, which in turn feeds back into further improvement of the GPT itself (Bresnahan & Trajtenberg, 1995, p. 84). Productivity gains therefore unfold through feedback loops rather than one-off technological shocks. However, this growth is fragile because it depends on coordination. Decentralized markets tend to underinvest in GPT-related innovation and complementary adoption, as benefits spill across actors Bresnahan & Trajtenberg (1995). Vertical and horizontal externalities are not fully internalized, causing market outcomes to fall short of the social optimum. Institutions therefore play a decisive role in enabling diffusion and coordinating complementary investments.

Evidence from the AI literature closely follows this pattern. Empirical work shows that AI adoption often produces delayed and uneven productivity effects, because gains depend on complementary changes in skills, workflows, and organizational structure. Using firm-level evidence, Brynjolfsson, Rock, and Syverson show that investments in AI are frequently accompanied by short-run adjustment costs, with productivity gains materializing only after firms undertake complementary investments in organizational capital and process redesign - a pattern they describe as a “productivity J-curve” (Brynjolfsson, Rock, & Syverson, 2021). This supports the GPT logic: improvements in AI increase the returns to downstream experimentation and application development, while successful applications raise demand for further advances in

data, compute, and models - making AI's economic impact contingent on how quickly and coherently these complements diffuse.

Taken together, these frameworks allow artificial intelligence to be understood as a Schumpeterian general purpose technology. Schumpeter provides the core logic of creative destruction, Aghion and Howitt formalize its growth and instability dynamics, Christensen explains firm-level displacement, and Bresnahan and Trajtenberg highlight the systemic and institutional conditions under which AI's economic impact materializes.

Part II: Swedish Growth School

The first part of this paper used creative destruction to explain why artificial intelligence is disruptive at its core - showing how AI operates through processes of replacement that reshape firms, reallocate tasks, and undermine existing technologies and organizational forms. This perspective helps clarify why AI generates pressure and displacement. What it leaves open, however, is how that disruption translates into concrete economic outcomes - across sectors, over time, and under real constraints. AI does not diffuse evenly across the economy - it enters some activities early while bypassing others, shifts tasks before organizations adapt, and reorganizes organizations before markets adjust. As a result, its economic effects appear uneven and delayed, producing pockets of high productivity alongside areas of slow adoption. Taken together, these patterns suggest that creative destruction captures the force of AI-driven change, but not the full process through which that force becomes economic transformation.

To understand this process, I will turn now to the Swedish Growth School. Their framework focuses on disequilibrium, complementarities, and coordination problems - allowing

AI to be examined not only as a disruptive innovation, but as a development process that depends on how technological change interacts with skills, institutions, and market structures. Rather than treating disequilibrium as a deviation from the norm, they extend the Schumpeterian work by placing structural imbalance at the center of economic transformation. Development unfolds through tensions that arise when new activities emerge alongside existing ones, and when technological possibilities move faster than organizations and institutions can adjust. As Erik Dahmén explains, “changes, being much of the essence of industrial dynamics, imply disequilibria which should not be called disturbances because they are essential in transformation processes” (Dahmén, 1988, p. 4). From this perspective, economic development is not best understood as a return to equilibrium after innovation shocks. It is an ongoing process of structural transformation - shaped by conflicts between old and new activities, and by gaps between what is technologically possible and what is economically organized.

Dahmén’s contribution is to make this perspective precise through the concept of development blocks. Growth depends on sequences of complementary changes that must move together - technology, finance, organization, skills, regulation, and markets. “The focus on complementarities and structural tensions makes the 'development block' concept fall into place. It refers to a sequence of complementarities which by way of a series of structural tensions, i.e. disequilibria, may result in a balanced situation.” (Dahmén, 1988, p. 5) When these complements do not align, the economy does not simply “grow slower.” Empirical work on AI adoption supports this view - evidence suggests that effective AI use depends on complementary investments in data infrastructure, organizational redesign, workforce skills, and governance arrangements, rather than on technological capability alone. Studies of firms further show that without changes in incentives, workflows, and decision-making structures, AI adoption often

remains fragmented and fails to translate into sustained productivity gains (Brynjolfsson, Rock, and Syverson, 2021).

Once development is framed as a block-building process, the role of disruption also looks sharper. Creative destruction is not just the disappearance of old firms or jobs. It is the pressure that emerges when new possibilities cannot be absorbed by existing structures. Dahmén (1988) calls attention to structural tensions - mismatches across stages of development that generate bottlenecks and conflicts. Those tensions are not merely obstacles; they are signals. They indicate unrealized potential and create incentives for re-coordination. This is a useful way to interpret current AI frictions. Skill shortages, uncertain liability regimes, uneven access to data and compute, organizational resistance, and slow diffusion are not simply adoption “failures.” They are predictable tensions inside an unfinished transformation.

Economic success of certain stages in a development process might require the realization of one or more specific complementary stages. This implies development potentials. It does not exclude, but does instead sometimes include, a depressive pressure in stages which are ‘premature’... The ‘premature’ stages thus could be stimulants of entrepreneurial initiatives. Even if dominated by depressive pressures a situation like this may release expansive forces. This represents a ‘structural tension’ (Dahmén, 1988, p. 5).

In other words, periods of slow adoption or apparent underperformance reflect not failure, but misalignment - signals that key complements are missing and that entrepreneurial coordination is needed before AI’s development potential can be realized.

While Dahmén's framework explains why transformation generates tensions inside an economy, it remains largely focused on sequences of complementarities. To understand how these complementarities are embedded in real-world technological architectures - and how coordination extends beyond individual sectors - we must widen from development blocks to technological systems. That widening is provided by Carlsson and Stankiewicz's technological systems perspective. Their argument is that innovation should be understood as a property of systems - made up of networks of firms, infrastructures, institutions, standards, and flows of knowledge - rather than as isolated inventions.

“A technological system is defined as a dynamic network of agents interacting in a specific economic/industrial area under a particular institutional infrastructure and involved in the generation, diffusion, and utilization of technology. Technological systems are defined in terms of knowledge/competence flows rather than flows of ordinary goods and services. In the presence of an entrepreneur and sufficient critical mass, such networks can be transformed into development blocks, i.e. synergistic clusters of firms and technologies which give rise to new business opportunities” (Carlsson & Stankiewicz, 1991, p. 93).

This shifts the focus of creative destruction from individual technologies to the systems in which they are embedded. Technologies do not simply replace one another - they reshape the interactions, alter interfaces, and reorganize entire value chains.

The implication is that coordination is not an automatic outcome of system structure - it is performed by entrepreneurs, who connect knowledge flows, assemble missing complements, and turn system-level possibilities into concrete business activity. For AI, this perspective matters

because the relevant object is not a single model, but an AI-enabled system - including data ecosystems, compute infrastructure, complementary software layers, organizational routines, and governance arrangements. The long-run economic impact of AI, therefore, depends on how well this system diffuses and how effectively its components are coordinated, including through non-market mechanisms such as education, regulation, and public infrastructure. Yet even this systems perspective leaves an important question open - who actually carries out the coordination that turns system-level potential into realized economic growth?

Acs and Sanders (2013) provide a useful framework by placing entrepreneurship at the center of an endogenous growth model. Their key move is to distinguish clearly between knowledge creation and knowledge commercialization. New knowledge - whether generated in corporate R&D laboratories, universities, or frontier AI research teams - does not translate into economic growth on its own. It becomes economically relevant only when entrepreneurs identify opportunities created by knowledge spillovers and act on them. Spillovers widen the space for entry and experimentation by making ideas and techniques available beyond their point of origin, but it is entrepreneurial commercialization that converts this latent potential into new products, new production processes, and new markets. This distinction is particularly important for AI, because much of its capability is inherently general-purpose. In such settings, the binding constraint is often not invention itself, but deployment and integration. Because knowledge spillovers are externalities, private incentives to commercialize do not fully align with social returns, which means that market equilibrium can tilt effort toward research while underproviding entrepreneurial exploitation. Even when the underlying technology is highly promising, growth can therefore be delayed or uneven. This logic maps directly onto contemporary AI debates, where economic value increasingly accrues to actors who successfully

commercialize, scale, and integrate applications, rather than to those who originally produced the underlying ideas. At the same time, this framework makes clear that entrepreneurship does not operate in isolation but depends on the surrounding institutional and organizational environment. (Acs and Sanders, 2013)

If technological systems define the structure within which innovation takes place, then the question of growth becomes a question of coordination - system-level potential does not turn into economic outcomes on its own but requires entrepreneurs and firms who can recognize opportunities created by technological change and actively connect missing complements. This shift from structures to agency brings a further question into focus: how market structure conditions which actors can innovate, and at which layers of the AI system. This is the question taken up by Acs and Audretsch through their industrial-organization perspective. Rather than treating innovation as a general property of firm size, they focus on the conditions under which different firms gain an advantage, arguing that the relevant issue is not whether large or small firms are more innovative in general, but “under which circumstances do large firms have the relative innovative advantage, and under which circumstances do small firms have the relative innovative advantage” (Acs & Audretsch, 1987, p. 573). The central point for Acs and Audretsch is that innovative outcomes depend on the structure of the markets in which firms operate. Innovative advantage varies with industry characteristics - including market concentration, barriers to entry, capital intensity, and the stage of the industry life cycle. Acs and Audretsch stress that the traditional Schumpeterian link between firm size and innovation does not hold across all contexts, noting that it is “not validated in every manufacturing industry,” and that innovation is shaped by “the size distribution of firms, the existence of certain barriers to entry, and the stage of the industry in the product life-cycle” (Acs & Audretsch, 1987, p. 570).

This helps explain why innovation tends to concentrate in different types of firms across different parts of an industry. In capital-intensive and highly concentrated settings, large firms often have an advantage because scale, financing, and appropriability matter. In more innovative and competitive environments, smaller firms can outperform because flexibility, specialized skills, and rapid experimentation are more important than scale. As Acs and Audretsch summarize, “large firms tend to have the relative innovative advantage in industries which are capital-intensive and concentrated,” while “small firms tend to have the relative innovative advantage in highly innovative industries” (Acs & Audretsch, 1987, p. 567).

To conclude, innovation does not simply adapt to existing market structures - it also reshapes them over time. Thus, this perspective points away from simple predictions for the future of AI effects. What we can assume right now is that it will, most likely, produce different innovation dynamics across the value chain - with concentration pressures in capital- and infrastructure-intensive layers, and greater contestability in downstream application and integration spaces. But it is important to stress that market structure, rather than firm size alone, becomes the key link between technological opportunity and innovative outcomes.

Part III: The Economic Impact of Artificial Intelligence

Up to this point, we have not treated AI as a single “thing” that arrives and changes the economy overnight. Instead, we have built a set of lenses - ways of seeing what AI is doing in economic terms. Creative destruction gives the core mechanism: AI matters because it works through replacement, reallocation, and the weakening of old arrangements from within. The Swedish Growth School then adds the missing layer: replacement alone does not tell us what the

economy becomes. Outcomes depend on whether the complements are built - skills, workflows, institutions, standards, infrastructure - and on whether someone actually coordinates them when they do not line up on their own. This part is where those lenses are used for their main purpose - the focus here is not on AI as a technical breakthrough, but on AI as a process of structural reorganization - and on what that process produces in practice. The question is no longer “can AI raise productivity?” but how and where it does, why the gains often arrive late, why they show up in pockets rather than everywhere at once, and why the same technology can generate both growth and tension at the same time. In other words, this section is about the economic outcomes - the real ones we can argue about, measure, and feel - not just the theoretical mechanism.

To keep the argument tight and readable, the analysis is organized around four domains where AI’s economic impact is most visible - and where the interaction between creative destruction and incomplete development blocks becomes easiest to trace. Productivity and growth come first, because AI affects output mainly through reorganization and complementary investment rather than instant gains. Labor markets and inequality follow, as task reallocation precedes job redefinition and can generate persistent mismatch when training and mobility lag. The third domain is firm dynamics and competition, where AI reshapes market structure across layers and where entrepreneurship coordinates missing complements to turn capability into markets. The final domain is institutions and policy, where spillovers and coordination failures make diffusion non-automatic and where governance determines the speed, breadth, and durability of AI’s economic gains. The main goal is not to claim one inevitable future for AI - it is to explain why AI’s economic impact is likely to be uneven - across sectors, across firms, and across time - and why that unevenness is not a side effect, but a structural feature of

transformation under creative destruction, especially when development blocks are still incomplete.

1) Productivity and growth

As AI systems are advancing quickly and now appear to be capable of performing tasks that were previously expensive or slow, many expect their economic impact to be immediate and direct. In this view, productivity should rise rapidly once the technology becomes available, while employment effects should also materialize quickly, whether in the form of large-scale job displacement or dramatic restructuring of work. These expectations differ in tone, but they share a common assumption: that the effects of AI should translate mechanically from technical capability to economy-wide outcomes (Brynjolfsson, Rock, & Syverson, 2021). However, viewed through the lenses developed earlier in this paper, that assumption becomes difficult to maintain. On the other hand, if AI is understood as a general purpose technology and, following Dahmén, as a development block still in the process of formation, neither immediate aggregate productivity gains nor sudden, uniform disruption should be the baseline expectation. General purpose technologies historically operate through gradual diffusion and through sequences of complementary changes, rather than through instant, economy-wide adjustment (Bresnahan & Trajtenberg, 1995). Framed this way, the central question is no longer whether AI is powerful, but how widely it is adopted and whether the surrounding complements are moving alongside it.

Recent data suggest that diffusion remains limited, even in economies commonly treated as being at the technological frontier. Using Census Bureau Business Trends and Outlook Survey data, the U.S. Congressional Research Service reports that the share of firms indicating AI use increased from 3.7 percent in September 2023 to 5.4 percent by February 2024, while also emphasizing that reaching workplace adoption rates of roughly 40 percent or more typically

takes over a decade (Congressional Research Service, 2024). This empirical fact already carries an important productivity implication - when adoption is partial and concentrated, productivity gains will also be partial and concentrated, appearing first in specific firms, teams, or functions rather than in aggregate statistics. Uneven productivity outcomes, in this sense, are not a puzzle but a direct consequence of uneven diffusion (Bresnahan & Trajtenberg, 1995)

Where productivity gains do appear, they can be understood through familiar Schumpeterian mechanisms. AI raises productivity by replacing parts of existing production systems, sometimes by automating discrete tasks, sometimes by improving the quality or speed of decision-making, and often by lowering the cost of cognitive work that previously required time and specialized labor (Acemoglu & Restrepo, 2020). In doing so, AI changes what counts as an efficient input at the task level. Activities such as customer interaction, software development, or internal coordination are reorganized rather than merely accelerated. Productivity gains therefore show up as replacement and reorganization, with competitive advantage shifting toward firms that can adapt their production systems more quickly. This is creative destruction operating within the firm (Schumpeter, 1942)

However, as the Swedish Growth School emphasizes, changes in production techniques alone are not sufficient to generate sustained growth. Dahmén's concept of development blocks highlights that growth depends on sequences of complementary changes that must move together, and that misalignment among these elements produces structural tensions rather than smooth adjustment (Dahmén, 1988). Applied to AI, this implies that technological readiness can coexist with organizational and institutional unpreparedness. Firms may have access to powerful models while still lacking integrated data systems, redesigned workflows, appropriate skills, or governance frameworks that allow AI to be deployed at scale. In such settings, AI's potential is

visible, but its economic impact remains limited. Firm-level evidence also aligns closely with this interpretation. The IMF's review of empirical studies conducted prior to the widespread diffusion of generative AI reports a wide range of estimated productivity effects from AI adoption, with impacts on sales per worker ranging from zero to approximately 6.8 percent depending on context (Comunale & Manera, 2024). This dispersion is not incidental. It reflects differences in complementary investments and organizational adjustment rather than differences in the underlying technology itself. AI definitely can raise productivity, but whether it does so depend on whether firms rebuild the surrounding system that makes the capability usable in day-to-day production - even when such rebuilding is underway, productivity gains need not appear immediately. Creative destruction is costly, and early stages of adoption are often dominated by experimentation, training, integration, and restructuring. Brynjolfsson, Rock, and Syverson formalize this dynamic as a productivity J-curve, arguing that general purpose technologies require large, often intangible complementary investments that are poorly measured, so early adoption can coincide with weak or even declining measured productivity before stronger gains materialize later (Brynjolfsson, Rock, & Syverson, 2021). From a Dahmén-style perspective, this reflects the cost of completing missing stages of a development block rather than a failure of AI to deliver economic value (Dahmén, 1988).

At the aggregate level, these dynamics help explain why firm-level gains do not automatically translate into rapid economy-wide growth. When production consists of complementary tasks, overall output can be constrained by bottlenecks in the hardest-to-adjust parts of the system. Chad Jones emphasizes that automating relatively easy tasks can generate large local gains while leaving aggregate productivity limited by weak links elsewhere in the production chain (Jones, 2023). Task-based macro calibrations by Acemoglu reinforce this point

quantitatively, suggesting that even under optimistic assumptions, aggregate total factor productivity gains from AI may be modest once bottlenecks and uneven task adjustment are taken seriously, with upper-bound estimates on the order of 0.66 percent over a decade (Acemoglu, 2024). Once these constraints are acknowledged, the expected macroeconomic pattern shifts. Rather than a permanent acceleration of growth rates, AI is more plausibly associated with level effects shaped by diffusion dynamics. The Penn Wharton Budget Model makes this explicit, projecting that generative AI could raise productivity and GDP levels by roughly 1.5 percent by 2035, with the largest contribution occurring during the diffusion phase and a smaller persistent lift thereafter (Penn Wharton Budget Model, 2025). This profile is consistent with a wave of restructuring driven by creative destruction, whose speed and reach depend on the completion of complementary investments. Seen through the general purpose technology lens, AI's productivity impact is therefore both real and fragile. A GPT does not generate growth simply by existing or by improving technically - it does so when complementary changes downstream reinforce its use and feed back into further innovation upstream. This is why recent projections increasingly emphasize adoption and absorption rather than raw capability alone (International Telecommunication Union, 2018). Whether AI delivers large economic gains depends less on how powerful the models become and more on whether firms, sectors, and institutions succeed in building the complementary structures that allow those capabilities to scale.

The productivity story that emerges is thus neither one of immediate transformation nor of inevitable disappointment - AI raises the ceiling of what is possible, but the economy still has to build the stairs. Those stairs consist of complementary investments, coordination, and time. When they move together, productivity gains can accumulate and spread. When they do not, AI's

power remains visible, but its economic impact appears delayed, uneven, and localized - exactly what one would expect from creative destruction unfolding within an incomplete development block.

2) Labor markets and inequality

If productivity is the headline outcome of AI adoption, labor markets are where its effects are actually lived. Many expect that if AI is powerful, it must either rapidly eliminate jobs or leave employment largely unchanged, and much of the anxiety around AI is driven by the fear of sudden displacement (Comunale & Manera, 2024). Both reactions, however, tend to miss the more structural point - creative destruction is rarely gentle, not because it necessarily destroys employment outright, but because it reshapes who is needed, which skills are rewarded, and which kinds of work feel stable over time.

AI also unsettles an older discussion on automation. Unlike earlier waves that were largely concentrated in routine manual tasks, AI reaches deeply into cognitive and professional workflows as well - drafting and summarizing text, assisting with coding, triaging customer support, reviewing documents, supporting diagnostics, scheduling, and administrative coordination (Acemoglu, 2024). This does not automatically imply mass unemployment, but it does imply faster task reallocation. And because jobs are bundles of tasks, faster task reallocation translates into more frequent reshaping of jobs rather than simple job loss. A useful way to describe what is happening - consistent with the logic developed earlier in this paper - is that AI shifts tasks before organizations adjust. In the short run, firms often deploy AI to rearrange tasks within existing roles rather than to eliminate roles altogether. Customer support workers increasingly supervise AI-based triage systems. Junior analysts become editors and validators of AI-generated drafts. Software engineers prototype and debug more quickly, while spending more

time on architecture, integration, and review. These changes can raise productivity, but they also change the content of work, and once the content of work shifts, the labor-market question becomes sharper: which skills become scarce, which become redundant, and which become newly valuable (PricewaterhouseCoopers, 2025)

This is where the Swedish Growth School lens becomes particularly important. Labor-market outcomes are not determined by “task exposure” alone, but by whether the complements that allow workers and firms to absorb change adjust quickly enough. These complements include training systems, credentialing and signaling mechanisms, hiring norms, mobility pathways, and institutional support for transitions (World Economic Forum, 2025). If these elements move in step with task reallocation, workers can shift into new roles and new task bundles, and the transition can be broadly beneficial even if it is disruptive. If they lag, the economy remains in disequilibrium longer than necessary. Workers are pushed toward new requirements without adequate scaffolding, firms face persistent mismatch between demanded and available skills, and the costs of adjustment appear as wage dispersion, vacancy pressure, and uneven opportunity across workers and regions (Economic Development Research Partners, 2020).

Much of the most informative recent evidence aligns more closely with this “job redesign and skills churn” channel than with a simple story of headcount collapse. PwC’s AI Jobs Barometer, for example, emphasizes that in AI-exposed occupations the content of work is changing rapidly, reporting that the skills sought by employers are evolving much faster in the most AI-exposed jobs than elsewhere (PricewaterhouseCoopers, 2025). Thus, if skill requirements rotate faster than education, training, and hiring systems can respond, then the labor complement is, by definition, lagging behind the technological stage, and structural tension is the predictable result (PricewaterhouseCoopers, 2025). Productivity gains can appear in pockets of

adoption while the aggregate labor market remains messy, because the binding constraint is not whether the technology works, but whether the economy can absorb a new way of organizing work.

This perspective also clarifies why inequality effects are plausible even in the absence of large, immediate unemployment. Creative destruction dismantles old skill bundles and occupational ladders while creating new ones, and the creation side tends to reward adaptability, complementary skills, and access to AI-enabled workflows. Workers and firms with stronger complements gain earlier, while those without them fall behind, even if no one is “replaced by AI” in a dramatic sense. Inequality can widen across workers, but also across firms, and firm-level inequality can feed back into labor inequality. Frontier firms are more likely to experiment, invest in training, redesign workflows, and capture early productivity gains, while lagging firms struggle to absorb new methods and therefore struggle to offer comparable wage growth or mobility (Szczepański, 2019).

The policy-oriented literature increasingly frames these dynamics as a distributional challenge as much as a growth challenge, which fits naturally with the lenses used here. A European Parliament briefing on the economic impacts of AI emphasizes that growth effects can be real while distributional pressures intensify, particularly if benefits concentrate in “superstar” firms and labor income fails to capture a proportional share of the gains (Szczepański, 2019). In Schumpeterian terms, this is not surprising. Replacement reallocates rents and bargaining power, and when a new system forms unevenly, gains are rarely distributed smoothly (Schumpeter, 1942). From a Swedish Growth School perspective, distributional outcomes are not merely after-effects; they are part of whether the development block completes. Weak labor absorption can

slow diffusion, raise resistance to adoption, and turn a productivity opportunity into a longer and more fragmented transition (Dahmén, 1988)

This lens also helps make sense of the wide range of “jobs at risk” claims without getting pulled into headline contests. Policy briefs often cite high exposure numbers, and the European Parliament review, for instance, summarizes evidence suggesting that a large share of jobs face some risk of computerization over time (EPRS, 2019. Szczepański, 2019). Read through the framework developed here, such statistics are not forecasts of unemployment - they are indicators of task composition and potential disruption, signaling that many jobs contain task bundles that could be reorganized. Whether that potential leads to displacement, redesign, or augmentation depends on complements: training capacity, labor mobility, organizational adaptation, and institutional support for transitions. Exposure captures technical substitutability; economic outcomes depend on adjustment.

The labor-market story that emerges is therefore neither “AI will take all the jobs” nor “AI will change nothing.” It is the more Schumpeterian, and more economic, claim that AI accelerates reallocation, reshapes the skill structure of work, and redistributes bargaining power and rents. As a result, wages and opportunities are likely to move unevenly even when headline employment remains relatively stable. The hard constraint, once again, is the Swedish Growth School one: how quickly labor-market complements can be built so that creation keeps pace with destruction, and so that productivity gains show up not only in output statistics, but in broad participation in the new structure of work.

3) Firm dynamics, entrepreneurship, and market structure

Firm dynamics are where creative destruction becomes concrete - much of the discussion around AI focuses on tasks and workers, but the deeper transformation takes place at the level of firms and competition. AI does not simply change how work is done - it changes what firms must be good at, which capabilities become scarce, and which business models remain viable. As a result, AI can produce both displacement and persistence at the firm level. Some firms will fail because their routines, incentives, and organizational structures lock them into older models. Others will survive because they are able to integrate AI into existing scale advantages, customer relationships, and distribution networks. At the same time, some entrants will succeed because they can build new products and markets faster than organizations can restructure. The point is not that AI systematically favors startups or large firms, but that it reshapes the structure of opportunity and constraint, and different actors occupy different positions within that structure (Schumpeter, 1942; Carlsson & Stankiewicz, 1991)

Christensen's theory of disruptive innovation helps clarify why displacement can be systematic even among well-managed firms. Organizations may be highly efficient at sustaining improvements to existing products and serving established customers, yet still struggle when the basis of competition shifts. AI-enabled opportunities often appear initially uncertain, low-margin, or peripheral to core markets, making them difficult to pursue within organizations optimized for incremental improvement. Even when organizations recognize the technology, they may fail to create the organizational space needed for serious experimentation. By the time the meaning of "performance" in the market shifts, capabilities that once generated advantage can become sources of lock-in rather than strength (Christensen, 1997). Displacement, in this sense, is not a failure of management quality but a structural outcome of how firms are organized to compete.

The Swedish Growth School adds a second, complementary layer to this account by reframing the role of entrepreneurs and new firms. Entrepreneurs are not only innovators; they are coordinators of missing complements (Acs & Audretsch, 1987). In the context of AI, this coordination role is unusually visible because so much economic value lies in translation and integration rather than in frontier invention alone. Many successful AI-oriented firms are not building new foundation models. Instead, they specialize in workflow integration, evaluation systems, data pipelines, governance and compliance tools, or sector-specific applications. Their contribution is to connect general-purpose capability to real organizational constraints, turning “a model that can do X” into “a system that reliably does X” within environments shaped by regulation, liability, budgets, and users who do not behave like benchmarks. In Dahmén’s terms, these firms advance the development block by linking stages that are technologically available but economically premature in isolation (Comunale & Manera, 2024).

This perspective aligns closely with Acs and Sanders’ distinction between knowledge creation and knowledge commercialization. In AI, spillovers are large. Methods diffuse quickly, open tools spread, workers move across firms, and downstream experimentation is constant. Yet spillovers do not become growth automatically. They become growth when firms and entrepreneurs commercialize them into products, processes, and markets (Acs & Sanders, 2013). In AI, commercialization often takes the form of organizational coordination rather than technical invention, and it is this coordination that converts system-level potential into realized economic activity (Acs & Sanders, 2013).

Market structure then shapes which actors can innovate at different layers of the AI system. The most plausible outcome is a layered landscape. Upstream layers - such as compute infrastructure, frontier model training, and large-scale data systems - are highly capital intensive

and exhibit scale economies, creating pressure toward concentration. Downstream layers - applications, workflow integration, and niche tools - can be more contestable, because specialized knowledge and rapid experimentation matter, and entry costs can be lower. However, contestability is not guaranteed. If upstream actors control access to critical inputs or restrict interfaces, downstream innovation can be constrained. In Swedish Growth School terms, this represents a coordination failure within the technological system, an imbalance that prevents the development block from fully completing (Carlsson & Stankiewicz, 1991)

The policy-oriented literature increasingly frames these dynamics through the lens of “superstar firms.” The concern is not simply concentration per se, but uneven diffusion. If frontier firms pull ahead while adoption and complementary investments lag elsewhere, aggregate productivity growth can appear disappointing even as cutting-edge capabilities advance rapidly (Szczepański, 2019). A European Parliament briefing explicitly highlights this channel, linking uneven diffusion of digital technologies to widening productivity gaps and potential winner-takes-all dynamics (Szczepański, 2019). This is one of the clearest points at which the two lenses used in this paper converge. Creative destruction explains the mechanism of reallocation and replacement at the firm level, while the Swedish Growth School explains why aggregate outcomes depend on whether complementary capabilities diffuse broadly enough to prevent the economy from fragmenting into high-productivity islands surrounded by a lagging periphery.

4) Institutions, coordination failures, and policy trade-offs

Institutions are not optional in this framework. If AI is understood as a general purpose technology and as a technological system, then a large share of its economic impact is shaped not only by technical capability but by coordination problems and spillovers. A simple way to put

this is that many of AI's gains spill across actors, while many of the costs remain local. A firm that invests in worker training creates skills that competitors can later hire away. A company that develops evaluation standards or auditing practices helps establish trust that benefits an entire sector. A hospital system that pioneers safe deployment protocols generates learning that diffuses well beyond its own boundaries. In each case, social returns exceed private returns, which means that even essential complements can be underprovided by markets acting on their own (Comunale & Manera, 2024). This is where the classic Aghion–Howitt logic on private versus social returns becomes practically relevant. Innovation often generates wedges between what is privately profitable and what is socially valuable, because spillovers are pervasive, appropriation is incomplete, and rents can be captured without fully internalizing broader benefits (Aghion & Howitt, 1992). In the context of AI, this wedge can appear as strong private incentives to invest in proprietary advantage, scaling, and performance, alongside weaker incentives to invest in complements that make diffusion safe, interoperable, and broadly accessible. Investments in auditability, standards, shared infrastructure, or worker transition systems are socially valuable, but they are often hard to appropriate privately, especially in competitive environments where benefits diffuse quickly.

Recent policy-oriented evidence reinforces this interpretation. The IMF's survey of AI's economic impacts emphasizes that measurable productivity and growth effects depend heavily on adoption and complementary investments, while governance remains fragmented across objectives such as competition, privacy, intellectual property, security, bias, and even financial stability (Georgieva, 2024). The key takeaway, viewed through the lenses of this paper, is not that regulation necessarily slows AI, but that governance itself functions as a complement. When governance is weak, inconsistent, or uncertain, diffusion can stall, or proceed in ways that

generate backlash, litigation, and later constraint, ultimately slowing the completion of the development block.

Acemoglu's recent work clarifies an additional trade-off that fits naturally within this framework. GDP growth is not the same as welfare if AI-driven expansion is fueled by costly investment surges or by the proliferation of socially unproductive or harmful tasks. He illustrates a scenario in which AI could raise GDP by roughly 2 percent while reducing welfare in consumption-equivalent terms, depending on which tasks and incentives dominate the deployment process (Acemoglu, 2024). Even without relying on the most provocative version of this claim, the implication is clear: institutions shape not only the rate of growth, but the quality of growth, influencing which types of innovation are rewarded, which complements are built, and who ultimately captures the gains.

This is also where distribution and public finance enter the picture without changing the analytical tone of the argument. If creative destruction reallocates income away from labor toward capital in certain sectors, or concentrates rents among a narrow set of firms, then tax bases, transfer demands, and fiscal capacity can shift even when aggregate GDP rises. A European Parliament briefing explicitly links AI-driven economic change to questions of taxation, redistribution, and the capacity of public systems to respond to widening disparities (Szczepański, 2019). In Swedish Growth School terms, institutions such as the welfare state, education systems, and labor-market policies are not external add-ons; they are part of the institutional complement set that determines whether structural tension resolves into development or hardens into prolonged conflict (Dahmén, 1988; Carlsson & Stankiewicz, 1991).

Framed this way, the policy question is best understood as one of complement-building rather than ideology. The issue is not “regulate AI” versus “do not regulate AI,” but which

institutional complements enable diffusion while managing risk and preventing lock-in. In practice, this includes clearer liability regimes for high-stakes deployment, standards for evaluation and auditing, procurement rules that support experimentation and scaling, education and training policies that speed labor-market adjustment, and competition and interoperability policies that prevent technological systems from becoming locked into narrow trajectories. These institutional choices do not sit outside the growth process; they shape whether AI's potential is translated into broad-based economic development or remains concentrated, fragile, and contested (Comunale & Manera, 2024).

Conclusion

This brings the argument to a full synthesis - the economic impact of AI is not written into the technology itself. AI intensifies creative destruction by operating through replacement and reallocation, but the direction and durability of its effects depend on how the surrounding system responds. What ultimately matters is not the speed of technical progress alone, but the pace at which missing complements are built and coordinated. At the macro level, AI behaves like a general-purpose technology whose effects diffuse unevenly and only materialize when complementary changes move alongside it. At the firm level, productivity gains emerge through reorganization rather than simple automation, favoring actors that can adapt their production systems. In labor markets, task reallocation precedes institutional adjustment, generating churn and distributional tension when training and mobility lag. At the system level, entrepreneurship, market structure, and institutions determine whether spillovers are translated into growth or dissipate into fragmentation.

The central divide, therefore, is not between optimism and pessimism about AI, nor between “more AI” and “less AI.” It is between completed and incomplete development blocks. When skills, organizations, markets, and institutions adjust in step, creative destruction can support broad-based development. When they do not, the same technology produces delayed, uneven, and contested outcomes. AI raises the ceiling of what is economically possible, but it does not determine how that ceiling is reached. That outcome depends on coordination - on whether complementary investments align, whether diffusion becomes systemic rather than local, and whether institutions function as enabling complements rather than sources of friction. In this sense, the economic impact of AI is not a technological inevitability. It is a structural result of how economies manage transformation under creative destruction.

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